

Shawn A. Burgess

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PROFESSIONAL SUMMARY:

- Senior Software Engineer specializing in modernization of multi-tenant scientific and enterprise platforms spanning chemistry, biology, registration, workflow, and data-intensive systems. Blend of deep hands-on full-stack delivery (C#, .NET, React, SQL, Java, Python) with architectural stewardship (service boundaries, API contracts, schema evolution, performance & scalability, identity & security). Known for converting legacy monolith & scripting surfaces into tested, observable services, instituting feature flag governance for safe incremental rollouts; and driving measurable reductions in operational friction (faster root cause isolation, lower regression rates, resilient bulk data processing).
- Core impact areas include: centralized chemistry & reaction processing, parent/child registration lineage, advanced searching, SAML SSO enablement, bulk import reliability (chem + bio sequences), internationalization & data privacy hardening, and CI/CD pipeline standardization (Azure DevOps + Kubernetes). Recognized as a trusted technical partner who accelerates delivery while elevating quality, observability, and maintainability.

STRENGTHS:

Full-Stack Web Engineering ▪ Systems & Data Architecture ▪ Application Modernization ▪ API & Integration Patterns ▪ Performance & Scalability ▪ Identity & Security (SAML SSO) ▪ DevOps & CI/CD Automation ▪ Kubernetes Deployments ▪ Legacy Remediation ▪ Observability & Diagnostics ▪ Cross-Functional Collaboration ▪ Technical Mentorship ▪ Process Improvement ▪ Quality Engineering

PROFESSIONAL EXPERIENCE:

Sr. Software Developer

January 2017 to June 2025

Arxspan / Bruker, Southborough and Billerica, Massachusetts

- Modernized multi-tenant ELN / registration platform: refactored core request & configuration services for extensibility and performance.
- Centralized chemistry processing: CDXML conversions, salt & stereochemistry normalization, stoichiometry & reaction APIs.
- Expanded registration workflows: parent/child lineage, gap-free numbering, stoichiometry grid, salts & structure property derivation, sequence validation.
- Engineered resilient bulk import & update pipelines (chem + bio sequences, SDF/SD, sub-data mapping, warning/error framework).
- Led SAML SSO & identity integration (enterprise metadata, multi-environment configs) streamlining customer onboarding.
- Built comprehensive Postman API test suite (multi-environment matrices + pipeline integration) raising regression confidence.
- Strengthened unit & integration test coverage; upgraded MSTest frameworks and introduced coverage tooling.
- Implemented feature flag strategy (stoichiometry, inventory linking, chemistry canvas, CAS / CRAIS checks) enabling incremental rollout.
- Introduced structured logging, performance instrumentation & interface log service improving diagnostics & mean time to recovery.
- Drove Azure DevOps + Kubernetes delivery improvements: templated pipelines, repeatable promotion, deployment reliability.

- Reduced technical debt across legacy Classic ASP / PHP / Python surfaces; migrated critical chem & search logic to services.
- Advanced internationalization (new locales, culture defaults) and data privacy (PII minimization).
- Refined permission & role gating (restricted search, inventory linking visibility) enhancing security posture.
- Mentored engineers in code review, observability patterns, and sustainable refactoring practices.

Sr. Application Developer

February 2013 to January 2017

The Hanover Insurance Company, Worcester, Massachusetts

- Engineered Java EE / Spring / Hibernate services & UI components powering core insurance workflows.
- Co-led git adoption: branching strategy, review discipline; improved release stability.
- Optimized ORM & SQL hotspots elevating responsiveness & reliability under load.
- Introduced structured logging & diagnostics accelerating root cause analysis.
- Extended Maven automation & dependency management, standardizing delivery.
- Mentored peers on refactoring strategy & maintainable design principles.

Sr. Technical Analyst

November 2008 to February 2013

EBSCO Publishing, Ipswich, Massachusetts

- Built internal workflow & content tools (Spring MVC, Hibernate, JSP, Bootstrap, jQuery) reducing manual editorial effort.
- Developed secure external content review portal (PHP, MySQL, JavaScript) enabling partner collaboration.
- Created ingestion / transformation utilities (XML, JSON, CSV) improving metadata consistency & search relevance.
- Implemented automated testing & structured build practices cutting regressions.
- Administered / tuned application servers; introduced monitoring & structured logging for faster resolution.
- Acted as technical mentor promoting clean architecture & sustainable refactoring.

Software Developer

March 2006 to September 2008

Compex Integrated Systems, Worcester, Massachusetts

- Developed and tested database software for a manufacturing environment.
- Worked with clients to deliver custom software to their specifications.

EDUCATION:

Bachelor of Science - Computer Science

May 2008

Worcester State College, Worcester, Massachusetts

TECHNICAL SKILLS

- **Languages:** C#, JavaScript / TypeScript, SQL, Java, Python, PHP, Classic ASP, HTML, CSS
- **Frameworks & Libraries:** .NET / ASP.NET MVC & Web API, React, Node.js, Spring, Hibernate, MSTest, xUnit, Bootstrap, jQuery
- **Cloud & DevOps:** Azure DevOps (Repos, Pipelines, Boards), Kubernetes, Docker, build & release templating, artifact/version management, environment promotion workflows, CI/CD
- **Data & Persistence:** SQL Server, relational modeling, migration scripting, bulk ingestion (SDF, CSV, XML, JSON)
- **Search & Analytics:** full-text search integration, structured & predicate search, performance & relevance tuning, export pipelines
- **Chemistry / Scientific Domain:** CDXML conversions, salts & stereochemistry normalization, stoichiometry & reaction data modeling, structure property hashing, sequence validation workflows
- **Security & Identity:** SAML SSO, role & permission gating, secure config & secrets handling, data privacy (PII minimization)
- **Quality Engineering:** Unit / integration testing, Postman multi-environment API suites, coverage instrumentation (Coverlet), static analysis (SonarQube), test data strategy
- **Observability & Diagnostics:** Structured logging, performance instrumentation, interface log services, error taxonomy & tracing
- **Performance & Reliability:** Hotspot profiling (ORM/SQL), pagination & batching strategies, resilient bulk import/update pipelines, defensive validation
- **Automation & Scripting:** Data migration & normalization scripts, indexing/reindexing utilities, health & monitoring scripts, build/deployment helpers
- **Modernization & Legacy Remediation:** Classic ASP / PHP to service + React transitions, incremental migration approaches, risk-managed refactors
- **Collaboration & Practices:** Code reviews, technical mentorship, Agile iteration planning, documentation, incremental delivery & rollout strategy
- **Internationalization:** Locale expansion, culture defaults, translation integration, user preference handling